

Performance Testing

Date	13 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	84.2%	0.81	0.79	0.80
Decision Tree	86.5%	0.84	0.83	0.83
Random Forest	91.3%	0.90	0.89	0.89
XGBoost	93.1%	0.92	0.91	0.91

XGBoost was selected as the best-performing model and saved for deployment based on the highest accuracy and F1-score across the test set.

Application Performance Testing

Test Scenario	Expected Result	Actual Result	Status
Single applicant prediction response time	< 2 seconds	~0.8 seconds	✔ Pass
Batch screening of 100 applicants	< 10 seconds	~6.4 seconds	✔ Pass
Concurrent requests (20 users)	No errors / crashes	All requests served successfully	✔ Pass
Invalid / missing input handling	Graceful error message	Form validation error shown	✔ Pass
Cloud endpoint (IBM Watson ML) latency	< 3 seconds	~1.9 seconds	✔ Pass

Load Testing Summary

⚡ Avg Response Time

0.8s (single) / 6.4s (100-batch)

👥 Concurrent Users Tested

20 simultaneous requests

✔ Error Rate

0% under tested load